

Virtual Museum AR/VR

AI-Powered Immersive Museum with WebXR, RAG & Personalized Curation

PHASE 0	PHASE 1	PHASE 2	PHASE 3	PHASE 4
Museum Foundation	RAG Core	XAI Gate	Personalization	API Wiring
■ Complete	■ Complete	■ Complete	■ Complete	■ Complete

Framework	Next.js 16.3.1 + React 19 + Three.js 0.185
Backend	FastAPI + Uvicorn + SentenceTransformers + Groq
XR Support	WebXR VR + AR via @react-three/xr 6.6.30
Galleries	3 galleries — Space, History, Natural History
Artifacts	15 curated artifacts with GLB 3D models
Core Research	Confidence-Aware Explainable RAG (threshold-gated LLM)

1. Project Overview

The Virtual Museum AR/VR is a full-stack, research-grade immersive museum experience that runs entirely in the browser. Visitors explore three-dimensional galleries, interact with 15 authentic cultural and scientific artifacts rendered as GLB 3D models, and receive AI-generated, strictly grounded answers from an AI Curator chatbot — all without installing any plugin or application.

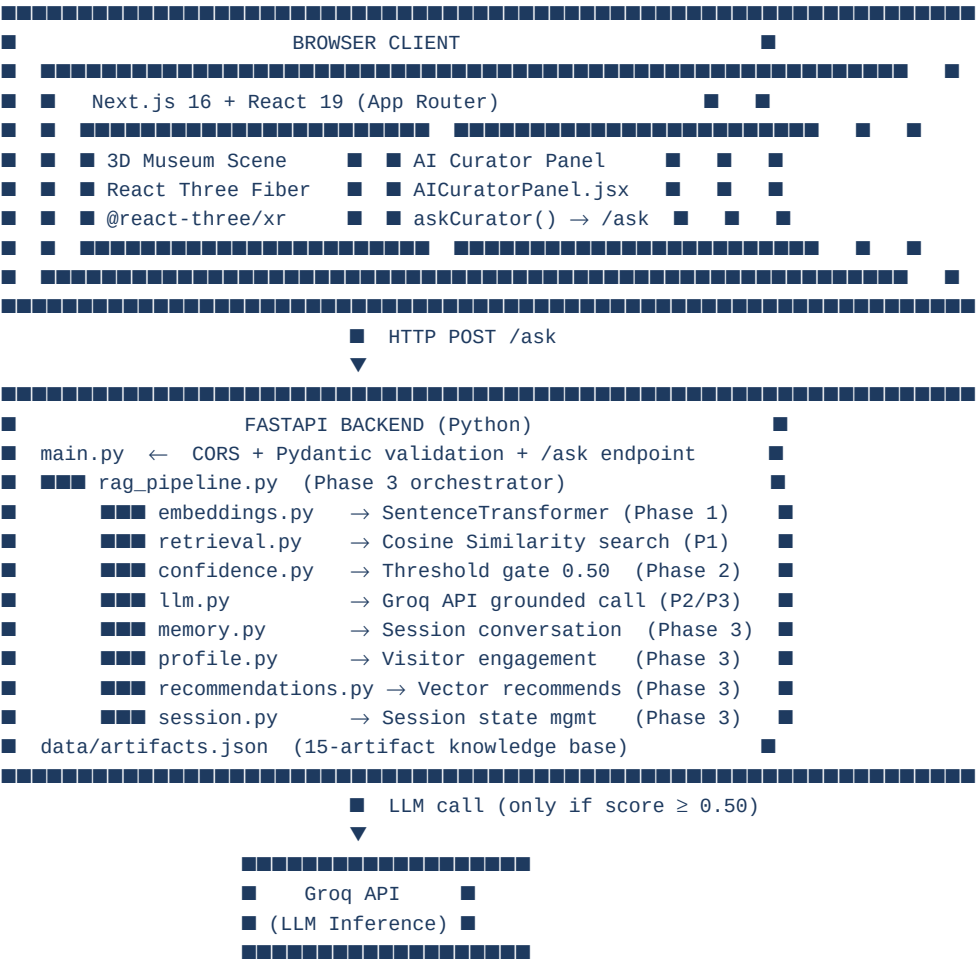
The core technical novelty is its Confidence-Aware Explainable RAG pipeline: before the LLM is ever called, a semantic retrieval similarity score gates the request against a configurable threshold (default: 0.50). Queries that do not sufficiently match the curated knowledge base are refused outright, preventing hallucination entirely. Every accepted answer is attributable to a specific artifact record with a visible confidence score.

2. User Journey

Step	Action	Technology
1	Visitor opens browser (desktop or headset)	Next.js SSR / CSR
2	3D museum loads — three gallery environments	Three.js + React Three Fiber
3	Visitor browses Space / History / Natural History galleries	3D scene navigation
4	Approaches artifact — Selection Beacon appears	Three.js proximity / raycasting
5	Clicks / gaze-selects artifact → Info Panel opens	React state + XR input
6	AI Curator Panel opens alongside artifact details	AICuratorPanel.jsx
7	Visitor types question, selects tone (Educational / Concise / Friendly)	React form
8	Next.js sends POST /ask to FastAPI backend	fetch API + JSON
9	Phase 1: question embedded + cosine similarity retrieval	SentenceTransformers + NumPy
10	Phase 2: Confidence gate — REFUSE if score < 0.50	confidence.py
11	Phase 3: Session memory + visitor profile updated	memory.py + profile.py
12	Phase 3: Groq LLM generates grounded answer (if accepted)	Groq API
13	Phase 3: 'You Might Also Like' recommendations computed	recommendations.py
14	JSON response rendered: score, answer, source, evidence, suggestions	AICuratorPanel.jsx

3. System Architecture

3.1 High-Level Architecture



3.2 RAG Pipeline Data Flow

Step	Component	Technology	Output
1	Visitor asks question	React state / form submit	Question string
2	HTTP request to backend	fetch POST → FastAPI /ask	HTTP 200 or error
3	Question embedding	SentenceTransformer all-MiniLM-L6-v2	384-dim vector
4	Cosine similarity search	NumPy (manual implementation)	15 similarity scores
5	Best artifact selected	argmax(scores)	artifact_id + score
6	Threshold gate	confidence.py is_relevant()	ACCEPT / REFUSE decision
7	Session memory load	memory.py format_history_for_prompt()	Last 3 turns text
8	LLM grounded call	Groq API + system prompt	Answer string

9	Visitor profile update	profile.py track_*	Analytics update
10	Recommendations	recommendations.py cosine_similarity()	Top-3 artifact list
11	JSON response → UI	FastAPI AskResponse model	Full structured payload

4. Technology Stack

4.1 Frontend

Technology	Version	Role
Next.js	16.3.1	React framework, App Router, SSR + CSR
React	19.2.8	UI component system, hooks, state management
React Three Fiber	9.7.0	Declarative Three.js rendering in React
Three.js	0.185.1	Core 3D rendering engine (WebGL)
@react-three/drei	10.7.8	Three.js helpers: GLTFLoader, OrbitControls
@react-three/xr	6.6.30	WebXR VR + AR session management
Tailwind CSS	4.x	Utility-first styling
Lucide React	1.33.0	Icon system (Bot, Send, Sparkles, etc.)

4.2 Backend (Python)

Technology	Version	Role
FastAPI	latest	REST API framework with automatic OpenAPI docs
Uvicorn	latest	ASGI web server for FastAPI
SentenceTransformers	latest	all-MiniLM-L6-v2 semantic embedding model
NumPy	latest	Vector math, cosine similarity, .npy cache
Groq SDK	latest	LLM grounded answer generation via Groq API
python-dotenv	latest	Environment variable management
Pydantic v2	latest	Request/response schema validation and typing

4.3 External Services & Models

Service / Model	Purpose
Groq API (xai-grok series)	LLM inference — strictly gated by confidence threshold
HuggingFace Hub	Auto-downloads all-MiniLM-L6-v2 on first backend start
all-MiniLM-L6-v2	384-dimensional sentence embedding model, ~80MB, CPU-compatible

5. Phase Breakdown

Phase 0 — Museum Foundation ■ Complete

The complete 3D museum experience including gallery environments, GLB model loading, artifact selection, WebXR VR/AR support, and the Artifact Info Panel.

Gallery	Artifacts (5 each)
Space Gallery	Apollo Lunar Module, Hubble Space Telescope, Mars Rover, Voyager Probe, ISS Module
History Gallery	Old Arrow Maker (Lewis), Benin Bronze, Egyptian Canopic Jar, Terracotta Warrior, Persian Rug
Natural History	T-Rex Skull, Ammonite Fossil, Blue Whale Skeleton, Woolly Mammoth Tusk, Smilodon Skull

Phase 1 — RAG Core Engine ■ Complete

Semantic retrieval system that identifies the best-matching artifact for any question using vector similarity, entirely without an LLM.

```
User Question
  ■
  ▼ SentenceTransformer (all-MiniLM-L6-v2)
Question Vector (384-dim)
  ■
  ▼ NumPy Cosine Similarity against 15 artifact vectors
Similarity Scores [score_0, score_1, ..., score_14]
  ■
  ▼ argmax(scores)
Best Artifact + Similarity Score + Evidence Text
```

Design Decision	Rationale
Manual cosine similarity (NumPy)	Full mathematical transparency — examinable formula for research review
all-MiniLM-L6-v2 model	384-dim compact model, fast, no GPU required, academically reproducible
.npy embedding cache	Prevents re-encoding on every server restart; auto-invalidated by artifact mtime
No LLM in Phase 1	Phase 1 is purely mathematical — retrieval is independent of any generative model

Phase 2 — Confidence-Aware Explainable RAG Gate ■ Complete

The research core of the project. The confidence gate prevents the LLM from being called for any query outside the curated knowledge base, eliminating hallucination at the architecture level.

```
RELEVANCE_THRESHOLD = 0.50

def is_relevant(similarity_score, threshold=0.50):
    return float(similarity_score) >= float(threshold)

# Gate logic in rag_pipeline.py:
if not is_relevant(score):
    return REFUSAL # LLM NOT CALLED
```

```
else:
    return call_groq_llm(artifact, question, tone)
```

Score Range	Decision	LLM Called?	UI Display
≥ 0.50	ACCEPT	■ Yes	Green score badge + answer + source + evidence
< 0.50	REFUSE	■ No	Red score badge + refusal message only

Note: The threshold value (0.50) was empirically tuned against the 15-artifact corpus. It is a cosine retrieval similarity score, not a calibrated statistical probability.

Phase 3 — Personalization Layer ■ Complete

Three independent sub-systems that operate after the Phase 2 gate and add session-awareness, tone-adaptive LLM responses, and vector-based exhibit recommendations.

Sub-system	File	Description
Session Memory	memory.py	Sliding window of last 3 conversation turns. Used ONLY for pronoun resolution (e.g. 'it', 'this exhibit').
Visitor Profile	profile.py	Anonymous telemetry: questions_asked counter, galleries engagement dict, artifacts_seen list. Z-score normalized.
Recommendations	recommendations.py	Artifact-to-artifact cosine similarity on Phase 1 vectors. Excludes current + seen artifacts. Returns top 3.
Tone Personalization	llm.py	Three styles: Educational (scholarly), Concise (2-3 sentences), Friendly (warm guide). Controls vLLM temperature.

Phase 4 — Frontend ↔ Backend Wiring ■ Complete

Full-stack integration connecting the Python RAG pipeline to the Next.js UI. Every pipeline output is surfaced to the visitor through the AI Curator Panel.

API Contract — POST /ask

```
// Request (Next.js → FastAPI)
{
  "session_id": "uuid-string-or-null",
  "question": "Who sculpted the Old Arrow Maker?",
  "artifact_id": "ART001",
  "tone": "educational"
}

// Response (FastAPI → Next.js)
{
  "answer": "The Old Arrow Maker was sculpted by Edmonia Lewis...",
  "source": { "artifact": "Old Arrow Maker", "gallery": "History Gallery" },
  "confidence": 0.83,
  "evidence": "Old Arrow Maker. Gallery: History Gallery. Category: American Neoclassical...",
  "refused": false,
  "session_id": "a1b2c3...",
  "tone": "educational",
  "related_suggestions": [
    { "artifact_id": "ART002", "artifact_name": "Benin Bronze",
      "gallery": "History Gallery", "similarity": 0.61 }
  ]
}
```

AI Curator Panel UI Features:

- Active Exhibit Context header — artifact name + ID badge
- Tone selector buttons — Educational / Concise / Friendly (with state highlight)
- Loading animation with Bot spinner while backend processes
- Confidence score badge — green for accepted, red for refused
- "Why this answer?" explainability toggle — explains cosine retrieval score
- Answer text box — styled red for refusal responses
- Source attribution section — artifact name + gallery (hidden on refusal)
- Evidence excerpt — verbatim text from curated artifact record

- "You Might Also Like" clickable recommendation cards — navigate to artifact

6. Key Technical Decisions & Research Rationale

Decision	Rationale
Manual cosine similarity in NumPy	Full transparency for research review — examinable one-line formula, no vector database black-box
LLM NOT called on refusal	Core research claim: threshold gating prevents hallucination on out-of-distribution queries at architecture level
Threshold = 0.50 (configurable)	Empirically tuned against 15-artifact corpus. Can be overridden per-request for ablation studies
all-MiniLM-L6-v2 embedding model	80MB, CPU-compatible, no GPU required. 384-dim vectors balance speed and semantic quality
.npy embedding cache	Avoids re-encoding 15 artifacts on every server restart. Cache auto-invalidates when artifacts.json is modified
Groq API (not OpenAI)	Lower latency inference; xai-grok models available; project API key pre-provided
Session memory window = 3 turns	Sufficient for pronoun resolution (e.g. 'it'); avoids prompt inflation and context window waste
Zero PII in visitor profile	GDPR-aligned anonymous telemetry; suitable for public museum deployment without consent screens
Strict LLM system prompt	Explicitly forbids outside knowledge, invented facts, dates, names. Conversation history cannot be treated as f
Evidence surfaced to UI	Explainability principle: visitor can verify which artifact record grounded the answer
Refusal message is deterministic	Same message every time — no LLM variation on refused queries — ensures predictable behavior

7. Project File Structure

```
virtual-museum/
├── src/
│   ├── app/                                # Next.js App Router pages
│   ├── components/
│   │   ├── curator/
│   │   │   ├── AICuratorPanel.jsx          # AI Curator UI (Phase 4)
│   │   │   ├── artifacts/                 # Artifact info panels
│   │   │   ├── museum/                   # 3D gallery scene components
│   │   │   ├── ar/                       # AR overlay components
│   │   │   ├── controls/                  # Camera / movement controls
│   │   │   ├── ui/                       # Shared UI components
│   │   └── data/
│   │       ├── artifacts.js               # 15-artifact frontend metadata
│   │       └── hooks/
│   │           ├── useWebXRSupport.js     # WebXR detection hook
│   │           └── utils/
│   │               ├── api/
│   │               └── curator.js         # askCurator() fetch client
│   └── backend/
│       ├── main.py                        # FastAPI app + /ask endpoint
│       ├── requirements.txt               # Python dependencies
│       ├── data/
│       │   ├── artifacts.json             # 15-artifact RAG knowledge base
│       │   └── rag/
│       │       ├── embeddings.py          # SentenceTransformer + cache (Phase 1)
│       │       └── retrieval.py            # Cosine similarity retrieval (Phase 1)
```

```
■      ■■■ confidence.py      # Threshold gate (Phase 2)
■      ■■■ llm.py            # Groq LLM grounded call (Phase 2/3)
■      ■■■ memory.py         # Session conversation memory (Phase 3)
■      ■■■ profile.py        # Visitor engagement profile (Phase 3)
■      ■■■ session.py        # Session state management (Phase 3)
■      ■■■ recommendations.py # Vector recommendations (Phase 3)
■      ■■■ rag_pipeline.py    # Phase 3 orchestrator
■
■■■ public/                  # Static assets, GLB 3D models
■■■ package.json             # Node.js frontend dependencies
■■■ next.config.mjs          # Next.js configuration
```

8. Running the Project

Backend (FastAPI)

```
cd virtual-museum/backend
venv\Scripts\activate          # Windows
pip install -r requirements.txt
uvicorn main:app --reload --port 8000
# → API available at http://localhost:8000
# → Swagger docs at http://localhost:8000/docs
```

Frontend (Next.js)

```
cd virtual-museum
npm run dev
# → Frontend at http://localhost:3000
```

Environment Variables

File	Variable	Value
backend/.env	GROQ_API_KEY	gsk_... (Groq API key)
backend/.env	FRONTEND_URL	http://localhost:3000
.env.local	NEXT_PUBLIC_BACKEND_URL	http://localhost:8000

9. Implemented Capabilities Summary

Capability	Implementation	Phase
3D immersive museum environment	Three.js + React Three Fiber	0
15 GLB artifact 3D models	GLTFLoader via @react-three/drei	0
WebXR VR mode	@react-three/xr immersive-vr session	0
WebXR AR mode + DOM overlay	@react-three/xr immersive-ar + DOM	0
Artifact selection beacon	Three.js proximity detection + raycasting	0
Semantic question embedding	all-MiniLM-L6-v2 SentenceTransformer	1
Cosine similarity retrieval	Manual NumPy implementation	1
Embedding cache (.npy)	Auto-invalidated by artifact mtime	1
Confidence gating (anti-hallucination)	threshold = 0.50 in confidence.py	2
Grounded LLM answers	Groq API with strict system prompt	2
Refusal for OOD queries	Deterministic refusal, LLM not called	2

Explainability: confidence score	Numerical score shown in UI	2
Explainability: source attribution	Artifact name + gallery in response	2
Explainability: evidence excerpt	Verbatim curated text shown in UI	2
Tone personalization (3 modes)	Educational / Concise / Friendly	3
Session conversation memory	Sliding window 3 turns, pronoun resolution	
Anonymous visitor profile	Questions, artifacts, galleries tracked	3
"You Might Also Like" engine	Artifact-to-artifact cosine similarity	3
FastAPI /ask endpoint	Pydantic validation + CORS	4
Full Next.js ↔ FastAPI wiring	AICuratorPanel.jsx + askCurator()	4
Structured JSON API contract	AskResponse Pydantic model	4

10. Remaining Work (Phases 5 & 6)

Phase	Name	Scope	Status
5	Evaluation Framework	Benchmark test suite; ablation study (grounded RAG vs raw LLM baseline); NoSQL core distribution analysis; pre-	Not started
6	Demo Polish	Attribution metadata UI; final deployment preparation; rehearsal walkthrough; High-end presentation polish	Not started